Comprehensive Scientific Investigation into Tsunami Physics, Mathematical Modelling, Operational Forecasting, and Causal Attribution

1. Research Mission, Scope, and Foundational Principles

Tsunamis are multi-scale fluid-dynamic phenomena initiated by transient, large-scale displacements of the ocean bed or water column. Developing an operational and scientifically defensible tsunami modeling and forecasting infrastructure mandates a unified interdisciplinary synthesis encompassing geophysics, physical oceanography, non-linear hyperbolic partial differential equation (PDE) theory, computational fluid dynamics (CFD), statistical data assimilation, machine learning, and high-performance software engineering.

This investigation establishes a mathematically rigorous, quantitatively grounded engineering blueprint for building a operational tsunami simulation system. The primary goal of this research is physical correctness and computational efficiency over superficial complexity or unvalidated predictive claims.

The physical life cycle of a tsunami can be categorized into four distinct computational regimes:

1. **Generation Phase:** Fluid volume displacement driven by transient boundary perturbations, including subduction megathrust ruptures, strike-slip faulting over sloping bathymetry, submarine mass failures, volcanic flank collapses, or atmospheric pressure waves.
2. **Open-Ocean Propagation Phase:** Trans-oceanic dispersion of long-period waves governed by shallow-water dynamics, Coriolis forces, and frequency dispersion.
3. **Shoaling and Coastal Interaction Phase:** Wave steepening, refraction, diffraction, reflection, and non-linear energy transfer as waves transition from deep abyssal plains ($h \approx 4000\text{ m}$) to shallow continental shelves ($h < 200\text{ m}$).
4. **Inundation and Coastal Run-up Phase:** Hydrodynamic bore formation, non-linear wave breaking, overland wetting/drying, bottom friction dissipation, and high-velocity fluid-structure interaction across complex coastal topography.

REGIME 1: GENERATION REGIME 2: DEEP-OCEAN PROPAGATION REGIME 3: SHOALING REGIME 4: COASTAL INUNDATION

Elastic/Mass Source Nonlinear/Dispersive Waves Wave Steepening Overland Flow & Bore Formation

[Okada / Tanioka Model] ---> [Linear / Dispersive Shallow Water] ---> [Nonlinear Shallow Water] ---> [Positivity-Preserving Wet/Dry]

2. Comprehensive Answers to Primary Research

Questions

Question 1: What physical mechanisms generate tsunamis, and how do their source mechanisms differ?

Tsunamis are generated by rapid volume displacements of the water column. The fundamental physical mechanisms include:

- **Subduction-Zone Megathrust Earthquakes:** Elastic rebound of the overriding plate causes massive vertical seabed elevation changes over thousands of square kilometers. Dip-slip motion dominates energy transfer.
- **Strike-Slip Earthquakes:** While slip is predominantly horizontal, strike-slip ruptures generate significant tsunamis when horizontal displacement acts across steep bathymetric gradients (e.g., continental slopes or trench walls), forcing vertical fluid movement.
- **Submarine Landslides and Mass Failures (SMFs):** Gravity-driven mass transport releases potential energy locally. Landslides act as continuous dipolar volume sources, generating short-wavelength, highly dispersive wave trains with extreme local run-up amplitudes but rapid radial decay.
- **Volcanic Flank Collapses, Explosions, and Caldera Subsidences:** Multi-phase mass entry or sub-surface explosive displacement generates localized, high-amplitude, non-hydrostatic wave fields.
- **Meteotsunamis and Bolide Impacts:** Atmospheric pressure anomalies moving at resonant phase speeds (Proudman resonance) or hypervelocity impact momentum transfers.

Question 2: Which governing equations describe tsunami dynamics across all phases?

Tsunami dynamics are modeled using hierarchical continuum mechanics systems:

- **Generation & Deep-Ocean Propagation:** Linear Shallow-Water Equations (LSWE) or weakly non-linear dispersive Boussinesq equations.
- **Shoaling, Coastal Run-up, and Inundation:** Two-Dimensional Non-Linear Shallow-Water Equations (NSWE) with bottom friction and topography terms.
- **3D / Local Mass Failure / Wave Structure Interaction:** Three-dimensional non-hydrostatic Navier-Stokes or multi-layer depth-integrated equations.

Question 3: Under what physical assumptions is each equation valid?

- **LSWE:** Valid when wave amplitude a is much smaller than ocean depth h_0 (\epsilon = a/h_0 \ll 1) and wavelength L is much greater than depth (\mu = (h_0/L)^2 \ll 1).
- **NSWE:** Valid under the hydrostatic pressure assumption, where vertical accelerations are negligible (\mu \ll 1), but finite-amplitude non-linear advection (\epsilon = O(1)) is critical.
- **Boussinesq Equations:** Valid when frequency dispersion is significant (\mu = O(10^{-2})) over long propagation distances, balancing non-linear steepening (\epsilon \approx \mu).
- **3D Non-hydrostatic Navier-Stokes:** Valid unconditionally across overturning wave breaking, vertical fluid accelerations, and multi-phase fluid flows, but computationally

expensive.

Question 4: Which numerical methods solve these equations accurately and efficiently?

High-resolution Finite Volume (FV) Godunov-type schemes employing exact or approximate Riemann solvers (e.g., HLL, HLLC) with Total Variation Diminishing (TVD) slope limiters provide the most stable and accurate methodology for hyperbolic shallow-water systems. They preserve mass conservation, resolve hydraulic jumps (bores) without unphysical oscillations, and handle wetting/drying boundaries stably.

Question 5: How can an earthquake rupture be translated into initial seafloor displacement and an initial tsunami condition?

Static elastic dislocation theory (Okada 1985 formulation) translates rectangular fault slip parameters into 3D crustal displacement fields. The vertical deformation u_z is augmented by the Tanioka and Seno (1997) correction for horizontal displacement across sloping bathymetry (\mathbf{u}_h \cdot \nabla b). The resulting seabed displacement is filtered through the Kajiura (1963) water-column transfer function 1/\cosh(kd_w) to produce the initial surface elevation \eta(x,y,0).

Question 6: How should bathymetry, coastal topography, tides, friction, dispersion and wetting/drying be represented?

Bathymetry and topography require unified Digital Elevation Models (DEMs) referenced to a common vertical datum (e.g., Mean Sea Level). Tides are added as dynamic background surface elevation offsets. Bottom friction is parameterised via space-varying Manning's n roughness formulations. Dispersion is incorporated via higher-order spatial derivative terms (Boussinesq). Wetting/drying is implemented via positivity-preserving cell reconstructions with explicit minimum depth thresholds (h_{dry} \approx 10^{-4}\text{ m}).

Question 7: Which data sources provide reliable observations for initialization, calibration and validation?

- **Source Initialization:** USGS ComCat seismic fault solutions, high-rate Global Navigation Satellite System (GNSS) surface displacements, and satellite InSAR deformation maps.
- **Propagation & Calibration:** NOAA DART ocean-bottom pressure (OBP) recorders and satellite altimetry tracks.
- **Inundation Validation:** UNESCO/IOC tide gauges, high-resolution post-event field survey run-up measurements, and SAR flood extent imagery.

Question 8: How can arrival times, wave heights, flow depths, inundation extents and run-up distances be calculated?

These metrics are extracted deterministically from time-dependent numerical grid outputs:

- *Arrival time (T_{arr}):* Time when local surface elevation exceeds a noise threshold

(\vert{}\eta\vert{} \ge 0.05\text{ m}).
- *Flow depth (d_f):* Local water depth over dry land (d_f = \max(0, \eta - b) for b > 0).
- *Run-up elevation (R_{max}):* Maximum vertical terrain height reached by the wetting front (b(x_{limit}, y_{limit})).
- *Inundation distance (L_{in}):* Maximum horizontal distance from the initial shoreline to the wetting front.

Question 9: Which inverse methods can estimate unknown tsunami sources from observations?

Tikhonov-regularized weighted least-squares inversions, Non-Negative Least Squares (NNLS), Bayesian Markov Chain Monte Carlo (MCMC) sampling, and Ensemble Kalman Filtering (EnKF) estimate finite-fault slip distributions by fitting observed DART pressure waveforms and coastal tide gauge records.

Question 10: How can uncertainty be propagated from earthquake source parameters through the entire simulation?

Uncertainty is propagated using Monte Carlo ensemble sampling across posterior distributions of source parameters (magnitude M_w, hypocentral depth, dip, strike, rake, and slip heterogeneity) to produce Probabilistic Coastal Hazard Maps with explicit confidence intervals.

Question 11: Which algorithms can meet near-real-time computational requirements?

Graphics Processing Unit (GPU)-accelerated 2D Finite Volume solvers (e.g., Tsunami-HySEA) achieve Faster-Than-Real-Time (FTRT) performance (simulating hours of propagation in minutes of wall-clock time). Precomputed scenario databases (e.g., NOAA SIFT) offer immediate linear wave lookup capabilities.

Question 12: Which machine-learning methods provide legitimate benefits, and where would their use be scientifically inappropriate?

Factorized Fourier Neural Operators (F-FNO) and DeepONets provide fast surrogate wave propagation emulators for real-time ensemble sampling. However, replacing non-linear shallow-water solvers with black-box neural networks in complex wetting/drying coastal inundation zones is scientifically inappropriate due to mass conservation violations and poor extrapolation beyond training distributions.

Question 13: How should the system distinguish an observed measurement, a numerical simulation, a probabilistic estimate and an official warning?

The software architecture must enforce strict schema separation and data provenance tagging. Observed measurements represent real-time physical sensor feeds (DART/tide gauges). Numerical simulations represent deterministic PDE solutions. Probabilistic estimates represent

ensemble statistics. Official warnings are administrative alerts issued strictly by authorized national/international warning centers (e.g., PTWC, NTWC).

Question 14: What minimum implementation would be scientifically credible, and what evidence would be required before making stronger claims?

A minimum credible implementation requires a 2D well-balanced Finite Volume NSWE solver passing all standardized NTHMP analytical and laboratory benchmarks (BP1 through BP6), initialized with an Okada dislocation source including bathymetric slope corrections. Stronger claims require independent post-event hindcast validation against field-surveyed inundation extents.

3. Tsunami Source Generation Dynamics and Earthquake Physics

Source Mechanisms and Physical Taxonomy

Tsunamigenic events originate from distinct geophysical mechanisms, each exhibiting fundamental differences in energy partitioning, spatial extent, frequency spectra, and initial displacement dynamics:

- **Subduction-Zone Megathrust Earthquakes:** Megathrust ruptures displace thousands of square kilometers of seafloor over rupture durations ranging from tens of seconds to several minutes. Dip-slip motion (reverse faulting) yields large vertical displacement fields.
- **Strike-Slip Earthquakes:** Historically considered low-risk tsunamigenic sources due to horizontal slip, strike-slip faulting generates significant tsunamis through horizontal displacement of steep bathymetric slopes.
- **Submarine Landslides and Mass Failures:** Submarine mass failures (SMFs) involve the gravity-driven motion of un-consolidated sediment blocks or debris flows along continental slopes. Landslide sources act as continuous dipolar volume fluxes. The initial wave frequency spectrum is significantly higher than that of seismic sources, leading to rapid radial dispersion.
- **Volcanic Collapses, Pyroclastic Flows, and Explosions:** Volcanic tsunamigenic mechanisms include subaerial flank collapses entering water bodies, underwater explosive eruptions, sub-surface caldera collapses, and dense pyroclastic density currents displacing surface water.
- **Bolide Impacts:** Extremely rare extraterrestrial impacts transfer kinetic energy into the water column through rapid displacement, atmospheric blast waves, and transient crater formation. Bolide sources create short-wavelength, highly dispersive waves with high initial amplitudes that decay exponentially with distance.

Elastic Dislocation Theory and the Okada Formulation

To translate earthquake fault parameters into seafloor deformation, static elastic dislocation theory in a homogeneous, isotropic half-space is used. The analytical formulation derived by Okada (1985) computes the explicit displacement field u_i(x_1, x_2, x_3) resulting from a

rectangular fault plane embedded at depth.
The fault plane geometry is defined by the following parameters:

- Fault length L (\text{m}) and width W (\text{m}).
- Depth to lower edge d (\text{m}) or centroid depth.
- Strike angle \theta (\text{deg}), measured clockwise from North (0^\circ \le \theta < 360^\circ).
- Dip angle \delta (\text{deg}), the angle between the fault plane and the horizontal surface (0^\circ \le \delta \le 90^\circ).
- Rake angle \lambda (\text{deg}), defining slip direction relative to strike line (\lambda=90^\circ for reverse slip, \lambda=-90^\circ for normal slip, \lambda=0^\circ for strike-slip).
- Total slip U (\text{m}), decomposed into strike-slip component U_1 = U \cos\lambda, dip-slip component U_2 = U \sin\lambda, and tensile opening component U_3.
- Elastic moduli: Lamé constants \lambda_e and shear modulus \mu_e, parameterized via the Poisson ratio \nu = \lam[span_29](start_span)[span_29](end_span)[span_49](start_span)[span_49](end_span)bda_e / (2(\lambda_e + \mu_e)) or the elastic ratio \alpha_e = (\lambda_e + \mu_e)/(\lambda_e + 2\mu_e).

The Cartesian coordinate frame aligns x along strike, y perpendicular to strike pointing in the dip direction, and z directed vertically downward (z \le 0).
The analytical solution for the displacement vector \mathbf{u} = (u_x, u_y, u_z)^T at the surface z=0 is given by the double-rectangle substitution:
u_i(x,y,0) = \frac{U}{2\pi} \left[\left[f_i(\xi, \eta) \right]_{\xi = x - L}^{\xi = x} \right]_{\eta = p - W}^{\eta = p}
where p = y \cos\delta + d \sin\delta, q =[span_32](start_span)[span_32](end_span)[span_52](start_span)[span_52](end_span) y \sin\delta - d \cos\delta, and R = \sqrt{\xi^2[span_85](start_span)[span_85](end_span) + \eta^2 + q^2}. The explicit function for dip-slip dislocation U_2 on vertical surface displacement u_z is expressed as:
u_z(x,y,0) = -\frac{U_2}{2\pi} \left[\frac{q d}{R(R+\xi)} + \sin\delta \arctan\left(\frac{\xi\eta}{q R}\right) - I_5 \sin\delta \cos\delta \right] \[span_33](start_span)[span_33](end_span)[span_53](start_span)[span_53](end_span)Bigg\vert{}_{\xi = x-L}^{\xi = x} \Bigg\vert{}_{\eta = p-W}^{\eta = p}
where I_5 is an elastic integrand function defined as:

I_5 = \begin{cases} \frac{\alpha_e}{\cos\delta} \left(\ln(R + d) - \sin\delta \ln(R + \eta) \right) & \text{i[span_34](start_span)[span_34](end_span)[span_54](start_span)[span_54](end_span)f } \cos\delta \neq 0 \\ -\frac{\alpha_e \eta}{R + d} & \text{if } \cos\delta = 0 \end{cases}

Horizontal Displacement and Bathymetric Slope Effects

Standard dislocation models compute only the vertical displacement of the seafloor \Delta z_{static} = u_z(x,y,0). However, in regions with significant bathymetric gradients (such as subduction trench slopes), horizontal motion of the sloping seafloor generates additional vertical ocean-water displacement, as demonstrated by Tanioka and Seno (1997).
The net effective vertical seafloor displacement \Delta h_{seafloor}(x,y) is derived by projecting the horizontal vector field (u_x, u_y) onto the local bathymetric bed slope \nabla b = (\frac{\partial b}{\partial x}, \frac{\partial b}{\partial y}), where b(x,y) is bed elevation measured positive upwards:

$$\Delta h_{seafloor}(x,y) = u_z(x,y) + u_x(x,y) \frac{\partial b}{\partial x} + u_y(x,y) \frac{\partial b}{\partial y} = u_z(x,y) + \mathbf{u}_[span_64](start_span)[span_64](end_span)[span_67](start_span)[span_67](end_span)[span_70](start_span)[span_70](end_span)h(x,y) \cdot \nabla b(x,y)$$

On steep continental slopes ($\nabla b \approx 0.1 \text{ to } 0.3$), the horizontal contribution $\mathbf{u}_h \cdot \nabla b$ can equal or exceed the direct vertical displacement u_z, making its inclusion mandatory for accurate near-field initialization.

Water Column Filtering and Sea Surface Perturbation

The initial free-surface elevation $\eta(x,y,0)$ does not instantly equal the seafloor deformation $\Delta h_{seafloor}(x,y)$. The intervening ocean water column acts as a low-pass spatial filter, damping high-wavenumber spatial components of seafloor motion. Under the linear potential flow assumption for an incompressible fluid of uniform depth d_w, the surface transformation is given in the spatial frequency domain by the Kajiura (1963) transfer function:

$$\tilde{\eta}(k_x, k_y, 0) [span_169](start_span)[span_169](end_span)= \frac{\Delta \tilde{h}_{seafloor}(k_x, k_y)}{\cosh\left(k d_w \right)}$$

where $k = \sqrt{k_x^2 + k_y^2}$ is the total spatial wavenumber. For long-wavelength seismic ruptures where $k d_w \ll 1$, $\c[span_170](start_span)[span_170](end_span)osh(k d_w) \approx 1$, validating the instantaneous long-wave static approximation:

$$\eta(x,y,0) \approx \Delta h_{seafloor}(x,y)$$

This approximation fails for localized, high-wavenumber features ($k d_w \ge 1$), such as landslide scarps or narrow fault ruptures, where spatial low-pass filtering must be applied to prevent numerical oscillations.

4. Governing Equations of Tsunami Dynamics

Conservation of Mass

Let $b(x,y,t)$ be the bed elevation positive upwards, $\eta(x,y,t)$ be the free-surface elevation relative to the undisturbed state ($z=0$), and $h(x,y,t)$ be the total water depth defined as:

$$h(x,y,t) = \eta(x,y,t) - b(x,y,t)$$

Let $\mathbf{u}(x,y,t) = (u, v)^T$ represent the depth-averaged horizontal velocity vector:

$$\mathbf{u}(x,y,t) = \frac{1}{h} \int_{b}^{\eta} \mathbf{U}_{3D}(x,y,z,t) \, dz$$

The exact depth-integrated continuous equation for conservation of mass is:

$$\frac{\partial h}{\partial t} + \nabla \cdot (h \mathbf{u}) = 0 \quad \implies \quad \frac{\partial \eta}{\partial t} + \frac{\partial (h u)}{\partial x} + \frac{\partial (h v)}{\partial y} = \frac{\partial b}{\partial t}$$

For static bathymetry ($\frac{\partial b}{\partial t} = 0$), this simplifies to $\frac{\partial \eta}{\partial t} + \nabla \cdot (h \mathbf{u}) = 0$.

Two-Dimensional Nonlinear Shallow-Water Equations (NSWE)

Under the hydrostatic pressure assumption ($\mu \ll 1$), vertical acceleration is neglected, and pressure distribution is purely hydrostatic: $p(z) = \rho g (\eta - z) + p_{atm}$. The 2D depth-integrated Non-Linear Shallow-Water Equations in conservative vector form are:

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}(\mathbf{U})}{\partial x} +$$

\frac{\partial \mathbf{G}(\mathbf{U})}{\partial y} = \mathbf{S}(\mathbf{U})
where the state vector \mathbf{U}, flux vectors \mathbf{F}(\mathbf{U}) and \mathbf{G}(\mathbf{U}), and source term vector \mathbf{S}(\mathbf{U}) are expressed in conservative variables:
\mathbf{U} = \begin{pmatrix} h \\ hu \\ hv \end{pmatrix}, \quad \mathbf{F}(\mathbf{U}) = \begin{pmatrix} hu \\ hu^2 + \frac{1}{2}g h^2 \\ huv \end{pmatrix}, \quad \mathbf{G}(\mathbf{U}) = \begin{pmatrix} hv \\ huv \\ hv^2 + \frac{1}{2}g h^2 \end{pmatrix} \mathbf{S}(\mathbf{U}) = \begin{pmatrix} 0 \\ -g h \frac{\partial b}{\partial x} + f_{cor} h v - \tau_{bx[span_97](start_span)[span_97](end_span)}/\rho \\ -g h \frac{\partial b}{\partial y} - f_{cor} h u - \tau_{by}/\rho \end{pmatrix}
The term f_{cor} = 2 \Omega \sin\phi_{lat} is the Coriolis parameter (\Omega = 7.2921 \times 10^{-5} \text{ rad/s}), and \boldsymbol{\tau}_b = (\tau_{bx}, \tau_{by})^T represents bottom shear stress. Bottom friction is parameterised via the quadratic Manning friction law:
\frac{\boldsymbol{\tau}_b}{\rho} = \frac{g n^2 \mathbf{u} \|\mathbf{u}\|}{h^{1/3}} = \frac{g n^2 \mathbf{u} \sqrt{u^2 + v^2}}{h^{1/3}}
where n is the Manning roughness coefficient (\text{s}\cdot\text{m}^{-1/3}).

Linearized Shallow-Water Equations (LSWE)

In deep-ocean environments where wave amplitude is small relative to ocean depth (\epsilon = a/h_0 \ll 1), non-linear advection (\mathbf{u} \cdot \nabla)\mathbf{u} and variable depth modifications h \approx h_0 are neglected. The governing system simplifies to the Linearized Shallow-Water Equations:
\frac{\partial \eta}{\partial t} + \nabla \cdot (h_0 \mathbf{u}) = 0, \quad \frac{\partial \mathbf{u}}{\partial t} + g \nabla \eta = 0
Combining these yields the non-dispersive 2D wave equation:
\frac{\partial^2 \eta}{\partial t^2} - \nabla \cdot \left(g h_0(x,y) \nabla \eta \right) = 0
The ocean wave speed (phase speed c and group velocity c_g) in the non-dispersive long-wave limit is:
c = c_g = \sqrt{g h_0}
This relation allows global travel-time calculations across deep ocean basins, but fails when frequency dispersion or finite-amplitude non-linear effects accumulate over thousands of kilometers.

Dispersive Boussinesq Formulations

When the wave spatial scale is moderate relative to water depth (\mu = (h_0/L)^2 = O(10^{-2})), frequency dispersion becomes significant. Higher-frequency wave components travel slower than long-wave components, causing wave trains to split into solitary wave packages. Non-dispersive NSWE fails to capture this phenomenon, requiring Boussinesq-type equations that retain O(\mu) terms.
The Madsen and Sørensen (1992) weakly non-linear, weakly dispersive Boussinesq formulation modifies the momentum equations by introducing higher-order spatial derivatives:
\frac{\partial (h \mathbf{u})}{\partial t[span_144](start_span)[span_144](end_span)} + \nabla \cdot \left(h \mathbf{u} \otimes \mathbf{u} \right) + g h \nabla \eta + \mathbf{\Psi}_{disp} = \mathbf{S}([span_145](start_span)[span_145](end_span)\mathbf{U})
where the dispersive vector term \mathbf{\Psi}_{disp} is parameterized as:
\mathbf{\Psi}_{disp} = - \left(B + \frac{1}{3} \right) h_0^3 \nabla \left(\nabla \cdot \frac{\partial

\mathbf{u}}{\partial t} \right) - B g h_0^3 \nabla \left(\nabla \cdot (h_0 \nabla \eta) \right)
where B = 1/15 is a tuning parameter optimized to match the linear dispersion relation of exact Airy wave theory up to k h_0 \approx 3.14:

\omega^2 = g k \tanh(k h_0)

Higher-Fidelity Formulations (3D Nonhydrostatic and Navier-Stokes)

In regions with steep bathymetry, vertical wall interactions, or multi-phase fluid flows (e.g., landslide generation or structural wave impacts), depth integration breaks down. Higher-fidelity models eliminate depth integration or restore vertical pressure variation:

- **Layered Nonhydrostatic Shallow-Water Models:** Divide the water column into N vertical layers, resolving vertical momentum by solving a 3D Poisson equation for non-hydrostatic pressure q(x,y,z,t).
- **Three-Dimensional Navier-Stokes (VOF/RANS/LES):** Solves full 3D incompressible Navier-Stokes equations coupled with Volume of Fluid (VOF) surface tracking for wave-structure interaction, coastal breaking, and structural loadings.

Dimensional Analysis and Consistency Verification

Checking physical dimensions across all primary terms ensures mathematical consistency:

- State Variable Depth h: [\text{L}] (\text{m}).
- Discharge Vector h\mathbf{u}: [\text{L}^2 \text{T}^{-1}] (\text{m}^2/\text{s}).
- Flux Gradient \nabla \cdot (h\mathbf{u}^2 + \frac{1}{2}gh^2): [\text{L} \text{T}^{-2}] (\text{m}/\text{s}^2).
- Bed Slope Source -gh\nabla b: [\text{L} \text{T}^{-2}] (\text{m}/\text{s}^2).
- Bottom Friction \tau_b/\rho = g n^2 u \vert{}u\vert{} / h^{1/3}: [\text{L} \text{T}^{-2}] (\text{m}/\text{s}^2), where n has dimensions [\text{T} \text{L}^{-1/3}] (\text{s}\cdot\text{m}^{-1/3}).

5. Numerical Methods, Discretization Schemes, and Algorithmic Frameworks

Finite Volume Godunov Methods and Riemann Solvers

Finite Volume (FV) formulations discretize the domain into control volumes C_i, expressing conservation laws in integral form:
\frac{d \mathbf{U}_i}{dt} + \frac{1}{|C_i|} \sum_{j \in N(i)} |\Gamma_{ij}| \mathbf{F}_{ij}^* = \mathbf{S}_i
where \mathbf{F}_{ij}^* is the numerical flux vector evaluated across boundary interface \Gamma_{ij} using a Riemann solver.
The HLLC (Harten-Lax-van Leer-Contact) solver restores the intermediate contact wave S_* missing from standard HLL schemes, resolving shear waves and material interfaces accurately:

S_* = \frac{S_L h_R (u_R - S_R) - S_R h_L (u_L - S_L)}{h_R (u_R - S_R) - h_L (u_L - S_L)}
\mathbf{F}_{HLLC}^* = \begin{cases} \mathbf{F}(\mathbf{U}_L) & \text{if } S_L \ge 0 \\ \mathbf{F}_*(\mathbf{U}_L) & \text{if } S[span_99](start_span)[span_99](end_span)_L < 0 \le S_* \\ \mathbf{F}_*(\mathbf{U}_R) & \text{if } S_* < 0 \le S_R \\ \mathbf{F}(\mathbf{U}_R) &

\text{if } S_R \le 0 \end{cases}

High-Resolution TVD Reconstruction and Limiters

Second-order spatial accuracy is achieved by reconstructing left and right state vectors \mathbf{U}_L, \mathbf{U}_R at cell interfaces using Total Variation Diminishing (TVD) slope limiters that suppress unphysical oscillations near shocks:
\mathbf{U}_{i+1/2, L} = \mathbf{U}_i + \frac{1}{2} \phi(r_i) (\mathbf{U}_i - \mathbf{U}_{i-1})
where r_i = \frac{\mathbf{U}_i - \mathbf{U}_{i-1}}{\mathbf{U}_{i+1} - \mathbf{U}_i} is the successive gradient ratio, and \phi(r) is the TVD limiter function (e.g., Monotonized Central, MC limiter):

\phi_{MC}(r) = \max\left(0, \min\left(2r, \frac{1+r}{2}, 2\right)\right)

Well-Balanced Property (C-Property Preservation)

A numerical scheme is *well-balanced* if it preserves the unperturbed quiescent equilibrium state (*lake-at-rest*) over arbitrary topography:
u = 0, \quad v = 0, \quad \eta = h + b = \text{Constant} \quad \implies \quad \nabla p = -g h \nabla b
Standard discretizations evaluate hydrostatic flux gradient terms \nabla (\frac{1}{2} g h^2) and bed slope source terms g h \nabla b separately, generating spurious numerical velocity spikes at steep bathymetric gradients. Well-balanced schemes balance these terms by reconstructing interface variables using surface elevation \eta rather than total depth h, hydrostatically adjusting interface depths:

h_{i+1/2, L} = \max\left(0, \eta_{i+1/2, L} - \max(b_i, b_{i+1})\right)

Wetting and Drying Boundary Treatment

Inundation modeling requires tracking moving shorelines across dry land (h=0). Unstable algorithms lead to negative depth calculations and numerical breakdown. Positivity-preserving wetting/drying schemes enforce an explicit depth threshold h_{dry} \approx 10^{-4} \text{ m} to govern cell transitions:

1. **Dry Cell Masking:** If h_i \le h_{dry}, cell i is designated dry, and horizontal velocity is set to zero (\mathbf{u}_i = \mathbf{0}).
2. **Interface Flux Inactivation:** Boundary fluxes across interface i+1/2 are zeroed out unless the adjacent wet cell surface elevation exceeds the dry cell bed elevation (\eta_{wet} > b_{dry}).
3. **Positivity Limiting:** Fractional time-steps update dry cell depth iteratively to ensure h_i^{n+1} \ge 0 unconditionally.

Courant-Friedrichs-Lewy (CFL) Stability Criterion

Explicit time integration (e.g., Strong Stability Preserving Runge-Kutta, SSP-RK2/RK3) is constrained by the multi-dimensional Courant-Friedrichs-Lewy (CFL) stability criterion:
\Delta t \le C_{CFL} \cdot \left[\min_{i,j}[span_108](start_span)[span_108](end_span) \left(\frac{|u_{i,j}| + \sqrt{g h_{i,j}}}{\Delta x} + \frac{|v_{i,j}| + \sqrt{g h_{i,j}}}{\Delta y} \right) \right]^{-1}
where C_{CFL} \le 0.5 for explicit 2D Finite Volume schemes. In deep ocean trenches (h \approx 10,000 \text{ m}), wave phase speed reaches \sqrt{g h} \approx 313 \text{ m/s},

imposing tight time-step restrictions across fine grid cells.

6. Domain Representation, Bathymetry, Topography, and Boundary Formulations

Digital Elevation Models and Horizontal/Vertical Datums

Tsunami models require continuous bathymetry-topography grids spanning from deep abyssal plains to inland run-up zones.

- **Primary Bathymetric Data Sources:**
  - *GEBCO (General Bathymetric Chart of the Oceans):* Global 15 arc-second grid (~450m resolution), suitable for deep-ocean propagation.
  - *NOAA NCEI Coastal Digital Elevation Models:* 1/3 to 1 arc-second grids (~10m–30m resolution), built for coastal hazard mapping.
  - *LiDAR / High-Resolution Topography:* Sub-meter to 3m grids, required to resolve seawalls, coastal dikes, and urban structures.
- **Datum Harmonization:** Reconciling bathymetric datums (typically referenced to Lowest Astronomical Tide, LAT, or Mean Lower Low Water, MLLW) with topographic datums (referenced to Mean Sea Level, MSL, or EGM96/ITRF ellipsoids) is mandatory. Failure to harmonize vertical datums creates artificial coastal scarps at the shoreline, disrupting inundation estimates.

Spatial Roughness Parameterization

Bottom friction dissipates wave energy during shoaling and overland flow. Roughness is parameterized via spatially varying Manning's n coefficients derived from land-use land-cover (LULC) datasets:

- *Smooth Deep Ocean Bed:* 0.015 - 0.020 \text{ s}\cdot\text{m}^{-1/3}
- *Coarse Sand / Gravel Shelf:* 0.025 - 0.030 \text{ s}\cdot\text{m}^{-1/3}
- *Paved Urban Areas / Roads:* 0.015 - 0.020 \text{ s}\cdot\text{m}^{-1/3}
- *Agricultural Land / Grassland:* 0.030 - 0.040 \text{ s}\cdot\text{m}^{-1/3}
- *Dense Coastal Forest / Mangroves:* 0.060 - 0.120 \text{ s}\cdot\text{m}[span_147](start_span)[span_147](end_span)^{-1/3}
- *High-Density Urban Infrastructure:* 0.080 - 0.150 \text{ s}\cdot\text{m}^{-1/3}

Domain Boundary Conditions

1. **Open-Ocean Transmission Boundaries:** Absorbing boundary conditions prevent artificial wave reflections at outer domain edges. The non-reflective Flather radiation condition matches incoming and outgoing characteristic variables:
   u_n = u_{n,0} \pm \sqrt{\frac{g}{h}} \left(\eta - \eta_0 \right)
1. **Sponge Layers:** Damping zones added near boundary margins apply Rayleigh friction coefficients to absorb outgoing wave energy without reflection.
2. **Solid Reflective Boundaries:** Enforced at dry land boundaries or domain walls using zero normal flow conditions: \mathbf{u} \cdot \mathbf{n} = 0.

7. Wave Transformation Kinematics, Shoaling, and Inundation Metrics

Shoaling Mechanics and Green's Law

In deep water, long wave amplitudes are small ($a \ll h$). As depth decreases, wave energy flux $F_e = E \cdot c_g$ is conserved. For non-dispersive long waves where $c_g = \sqrt{g h}$ and total energy density $E = \frac{1}{2} \rho g a^2$, conservation of energy flux across a channel of width b_w yields Green's Law:

$$F_e = \frac{1}{2} \rho g a^2 \sqrt{g h} \cdot b_w = \text{Constant} \quad \implies \quad \frac{a}{a_0} = \left(\frac{h_0}{h} \right)^{1/4} \left(\frac{b_{w,0}}{b_w} \right)^{1/2}$$

As water depth decreases from $h_0 = 4000 \text{ m}$ to $h = 10 \text{ m}$, wave amplitude amplifies by a factor of $(4000/10)^{1/4} \approx 4.47$. Concurrently, horizontal wave speed drops, shortening wavelength $L = c \cdot T$ and increasing wave steepness until non-linear wave breaking or run-up occurs.

Precise Inundation Metrics and Definitions

To avoid ambiguity in forecast communication, hydrodynamic quantities are strictly defined:

- **Free-Surface Elevation (η):** Instantaneous water surface height relative to fixed vertical geoid datum ($z=0$, e.g., MSL) (m).
- **Wave Amplitude (a):** Maximum vertical deviation of free surface above background ocean level (m).
- **Flow Depth (d_f):** Local vertical height of water column above local ground surface: $d_f(x,y,t) = \max\left(0, \eta(x,y,t) - b(x,y)\right)$ for $b(x,y) > 0$ (m).
- **Maximum Run-Up Elevation (R_{max}):** Maximum vertical elevation above reference datum reached by water at its furthest inland extent (m).
- **Maximum Inland Inundation Distance (L_{in}):** Maximum horizontal distance penetrated by water shoreward from pre-event coastline (m).
- **First Wave Arrival Time (T_{arr}):** The time at which water surface displacement at a station exceeds a defined physical threshold ($\vert{}\eta(t)\vert{} \ge \eta_{thresh}$, typically $\eta_{thresh} = 0.05 \text{ m}$).

8. Inverse Source Estimation and Data Assimilation Frameworks

Mathematical Formulation of Source Inversion

Let $\boldsymbol{\theta} \in \mathbb{R}^m$ be the vector of unknown source parameters (e.g., fault sub-block slip amplitudes U_k), and $\mathbf{y} \in \mathbb{R}^n$ be the vector of time-series observations. The forward physical model is operator $\mathcal{H}(\boldsymbol{\theta})$:

$$\mathbf{y} = \mathcal{H}(\boldsymbol{\theta}) + \boldsymbol{\varepsilon}$$

where $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{C}_\varepsilon)$ represents

measurement and model noise.
When \mathcal{H} is linear (e.g., linear long-wave propagation combined with Okada dislocation), the system simplifies to matrix operator \mathbf{G}:
\mathbf{y} = \mathbf{G} \boldsymbol{\theta} + \boldsymbol{\varepsilon}
where G_{ij} is the Green's function representing wave response at sensor i produced by unit slip on sub-fault j.

Regularized Least-Squares Inversion

Due to limited ocean-sensor coverage, \mathbf{G} is ill-conditioned. Tikhonov regularization is applied to stabilize the inverse problem:
\hat{\boldsymbol{\theta}} = \arg\min_{\boldsymbol{\theta}} \left((\mathbf{y} - \mathbf{G}\boldsymbol{\theta})^T \mathbf{C}_\varepsilon^{-1} (\mathbf{y} - \mathbf{G}\boldsymbol{\theta}) + \lambda^2 \|\mathbf{L} \boldsymbol{\theta}\|^2 \right)
where \mathbf{L} is a discrete Laplacian smoothing operator enforcing spatial smoothness across adjacent fault segments, and \lambda is the regularization parameter determined via L-curve optimization. Non-negativity constraints (\theta_j \ge 0) prevent unphysical back-slip.

Sequential Data Assimilation (Ensemble Kalman Filter)

For continuous wavefield updating, the Ensemble Kalman Filter (EnKF) assimilates ocean-bottom pressure and DART measurements sequentially without requiring re-inversion of source parameters. The state vector \mathbf{x}_k = (\boldsymbol{\eta}_k, \mathbf{u}_k, \mathbf{v}_k)^T at time k is tracked across N_e ensemble members:
\mathbf{x}_{k|k}^{(i)} = \mathbf{[span_204](start_span)[span_204](end_span)x}_{k|k-1}^{(i)} + \mathbf{K}_k \left(\mathbf{y}_k^{(i)} - \mathbf{H}_k \mathbf{x}_{k|k-1}^{(i)} \right) \mathbf{K}_k = \mathbf{P}_{k|k-1}^f \mathbf{H}_k^T \left(\mathbf{H}_k \mathbf{P}_{k|k-1}^f \mathbf{H}_k^T + \mathbf{R}_k \right)^{-1}
where \mathbf{P}^f is forecast covariance, \mathbf{R} is measurement noise covariance, and \mathbf{K} is the Kalman gain matrix.

9. Operational Forecasting Architectures and Real-Time Systems

Operational forecasting mandates balancing physical model fidelity with strict latency constraints. Precomputed scenario databases (e.g., NOAA SIFT/RIFT) scale and linearly superpose precomputed unit source waveforms, offering rapid lookup capabilities (\approx 1 - 10 \text{ seconds}). However, database matching fails for complex rupture geometries, landslides, or localized inundation dynamics.
Conversely, Faster-Than-Real-Time (FTRT) GPU On-the-Fly Simulations (e.g., Tsunami-HySEA) solve fully non-linear shallow-water equations directly upon receipt of earthquake parameters. Multi-GPU systems compute 8 to 24 hours of wave propagation and coastal inundation in under 2 minutes wall-clock time, resolving dynamic time-dependent sources and non-linear coastal shoaling without pre-computed structural assumptions.

10. Physics-Informed Machine Learning and Neural

Operator Surrogates

Factorized Fourier Neural Operators (F-FNO) learn mappings between continuous function spaces in the spectral domain. By parameterizing integral kernel operators directly in Fourier space, F-FNOs solve wave propagation PDEs across grid resolutions without retraining:
v_{l+1}([span_231](start_span)[span_231](end_span)[span_233](start_span)[span_233](end_span)[span_235](start_span)[span_235](end_span)x) = \sigma \left(W v_l(x) + \mathcal{F}^{-1} \left(R_{\phi} \cdot \mathcal{F}(v_l) \right)(x) \right)
where \mathcal{F} is Fast Fourier Transform, R_{\phi} is learnable complex tensor weights truncating high Fourier modes, and W is pointwise linear transformation. Factorized FNOs separate 2D spatial spectrum calculations, reducing memory footprints by 70\% and accelerating basin-scale emulator runtime to
[span_300](start_span)[span_300](end_span)\approx 8.5 - 12 \text{ seconds} per scenario.
Machine learning surrogates offer fast offshore wave propagation emulation. However, replacing non-linear shallow-water solvers with black-box neural networks in dry-bed coastal inundation zones is scientifically inappropriate due to mass conservation violations and poor extrapolation beyond training distributions.

11. Uncertainty Quantification and Probabilistic Hazard Assessment

Uncertainties are categorized as Aleatory (inherent randomness in fault slip heterogeneity) or Epistemic (knowledge limits in magnitude estimation \pm 0.2 M_w, bathymetric DEM noise, and Manning friction parameterization).
Probabilistic Coastal Hazard Analysis (PCHA) constructs ensemble realizations by drawing source parameters from posterior probability distributions obtained via seismic inversions:
\boldsymbol{\theta}^{(k)} \sim P(\boldsymbol{\theta} \mid \mathbf{y}_{realtime}), \quad k = 1, \dots, N_e
For each member realization \boldsymbol{\theta}^{(k)}, forward simulation generates localized flow depth fields d_f^{(k)}(x,y). The exceedance probability for a threshold depth d^* (e.g., d^* = 2.0 \text{
m}[span_208](start_span)[span_208](end_span)[span_212](start_span)[span_212](end_span)) is evaluated as:
P(d_f(x,y) \ge d^*) = \frac{1}{N_e} \sum_{k=1}^{N_e} \mathbb{I}\left(d_f^{(k)}(x,y) \ge d^* \right)
where \mathbb{I}(\cdot) is the indicator function. The resulting output yields probabilistic hazard maps defining confidence bands (5\%, 50\%, 95\% bounds) rather than a single deterministic inundation line.

12. Authoritative Datasets, Observations, and Infrastructure

Robust simulation and operational validation rely on authoritative global and regional data systems:

- **GEBCO 2024 Grid:** Global 15 arc-second ocean bathymetry grid accessed via Direct HTTPS / NetCDF for deep ocean wave propagation.

- **NOAA NCEI Coastal DEMs:** 1/3 to 1 arc-second elevation models accessed via REST API / Web Coverage Service for coastal inundation domains.
- **NOAA DART Array:** Real-time 15-second deep-ocean pressure recorder network accessed via NDBC API for wave verification.
- **IOC Sea Level Station Network:** Global 1-minute coastal tide gauge network accessed via IOC SLSM REST API for arrival time validation.
- **USGS ComCat Earthquake API:** Real-time seismic fault parameters accessed via GeoJSON REST API for source initialization.
- **FES2014 Global Tidal Model:** Global 1/16-degree tidal range and harmonic amplitudes accessed via Copernicus Marine API for tidal de-tiding.

13. Verification Protocols, Validation Metrics, and Benchmark Suite

To ensure numerical stability, accuracy, and reproducibility, models must pass standard validation suites established by the National Tsunami Hazard Mitigation Program (NTHMP):

1. **Analytical Benchmark 1 (BP1):** Solitary wave run-up on a planar beach against Carrier-Greenspan analytical solutions.
2. **Laboratory Benchmark 4 (BP4):** Wave run-up on a conical island testing 2D radial wave shoaling and diffraction.
3. **Laboratory Benchmark 6 (BP6):** Monai Valley Okushiri Island inundation testing 3D complex coastal topography run-up.
4. **Field Benchmarks:** 2011 Tohoku and 2010 Maule historical hindcasts evaluating basin propagation, DART pressure records, and field inundation maps.

Model skill is quantified using normalized statistical metrics:

- *Wave Height RMSE:* RMSE = \sqrt{ \frac{1}{N} \sum_{i=1}^N \left(\eta_{sim}(t_i) - \eta_{obs}(t_i) \right)^2 }
- *Arikawa Inundation Intersection over Union (IoU):* IoU = \frac{\text{Area}\left(\Omega_{sim} \cap \Omega_{obs} \right)}{\text{Area}\left(\Omega_{sim} \cup \Omega_{obs} \right)} (Acceptable skill threshold IoU \ge 0.70).
- *Willmott Skill Score:* d_{skill} = 1 - \frac{\sum_{i=1}^N \left\vert{} \eta_{sim}(t_i) - \eta_{obs}(t_i) \right\vert{}^2}{\sum_{i=1}^N \left(\left\vert{} \eta_{sim}(t_i) - \bar{\eta}_{obs} \right\vert{} + \left\vert{} \eta_{obs}(t_i) - \bar{\eta}_{obs} \right\vert{} \right)^2}

14. Critical Requirement: Tsunami Detection, Classification, and Causal Attribution

Source Classification and Ground Deformation vs. Water Change

Differentiating genuine tsunami waves from background oceanographic noise, atmospheric surges, or non-tsunamigenic seismic events requires multi-sensor causal attribution. Ground-deformation measurements (GNSS, InSAR) alone cannot confirm a tsunami. Static ground deformation proves fault slip occurred, but does not confirm fluid displacement. For instance, deep focus earthquakes (depth > 100\text{ km}) or continental strike-slip ruptures produce measurable GNSS land displacement without displacing water columns. Direct oceanic

water-column observations (DART, tide gauges) are mandatory to verify wave generation.

Signal Processing Pipeline

Raw ocean sensor time series contain superimposed signals: \eta_{obs}(t) = \eta_{tide}([span_189](start_span)[span_189](end_span)[span_193](start_span)[span_193](end_span)t) + \eta_{surge}(t) + \eta_{seiche}(t) + \eta_{tsunami}(t) + \varepsilon_{noise}(t). Differentiating tsunami waveforms uses a sequential processing pipeline:

1. **Harmonic De-Tiding:** Real-time harmonic analysis (UTide / FES2014) removes astronomical tidal components.
2. **Bandpass Filtering:** Tsunami signals occupy period bands between 2 \text{ min} \le T \le 120 \text{ min}. A Butterworth bandpass filter suppresses low-frequency weather trends (T > 2\text{ hours}) and high-frequency wind waves (T < 2\text{ min}).
3. **Continuous Wavelet Transform (CWT):** Identifies transient energy concentration in time-frequency space using the Morlet wavelet:

W(a, b) = \frac{1}{\sqrt{a}} \int_{-\infty}^{\infty} \eta_{detided}(t) \psi^*\left(\frac{t-b}{a}\right) dt
RAW SENSOR SIGNAL ---> [UTide Harmonic De-tiding] ---> [Butterworth Bandpass Filter] ---> [Morlet Wavelet CWT] ---> ISOLATED TSUNAMI SIGNAL

Bayesian Causal Decision Engine

A Bayesian decision framework evaluates the likelihood ratio \Lambda across competing hypotheses (H_1: Genuine Tsunami vs. H_0: Meteorological / Sensor Noise):

\Lambda = \frac{P(\mathbf{y}_{DART}, \mathbf{y}_{gauge} \mid H_1, \mathbf{\theta}_{seismic})}{P(\mathbf{y}_{DART}, \mathbf{y}_{gauge} \mid H_0)} \Lambda = \frac{\exp\left(-\frac{1}{2} (\mathbf{y} - \mathbf{y}_{sim})^T \mathbf{C}_\varepsilon^{-1} (\mathbf{y} - \mathbf{y}_{sim})\right)}{\exp\left(-\frac{1}{2} \mathbf{y}^T \mathbf{C}_{noise}^{-1} \mathbf{y}\right)}

Five-Level Evidence Classification Schema

1. **Level 0 (Insufficient Data):** Seismic alert received, but no ocean sensors within propagation range.
2. **Level 1 (No Evidence Detected):** Nearby DART buoys display flat de-tided signals (\vert{}\eta\vert{} < 0.01\text{ m}).
3. **Level 2 (Possible Tsunami):** Seismic parameters meet magnitude/depth thresholds (M_w \ge 6.5, depth \le 50\text{ km}), but wave signal unconfirmed on deep buoys.
4. **Level 3 (Multiple Observations Consistent):** DART buoys and GNSS displacement vectors confirm matching wave period, phase, and arrival time predictions.
5. **Level 4 (Confirmed by Authoritative Source):** Event confirmed by official authorities (PTWC/NTWC).

15. System Software Engineering Architecture and Implementation Workflow

Architectural Comparison of Scientific Tsunami Codes

- **GeoClaw (Clawpack Suite):** Solves 2D NSWE using Finite Volume Godunov schemes on adaptive mesh refinement (AMR) grids. Parallelized via OpenMP / MPI on CPUs. Excellent for multi-scale domain resolutions, but execution speed is slower than GPU solvers.
- **Tsunami-HySEA (EDANYA Group):** Solves 2D NSWE using well-balanced high-order Finite Volume methods on GPU architectures (CUDA / SYCL). Achieves Faster-Than-Real-Time (FTRT) speeds (>100x real time), making it optimal for operational forecasting systems.
- **MOST (NOAA):** Solves 2D NSWE using Finite Difference splitting schemes on structured nested grids. Standard operational code used by NOAA warning centers.

End-to-End Operational Workflow Algorithm

Algorithm: End-to-End Real-Time Tsunami Forecasting Workflow

Input : USGS Seismic Alert (Lat, Lon, Depth, Mw, Strike, Dip, Rake)
Output : Inundation H[span_218](start_span)[span_218](end_span)[span_223](start_span)[span_223](end_span)[span_228](start_span)[span_228](end_span)azard Map, Travel Time Field, Probabilistic Hazard Bounds

1. Ingest seismic event parameters via USGS ComCat GeoJSON API.
2. Extract spatial bathymetry/topography domain from unified NetCDF DEM dataset.
3. Compute 3D crustal displacement vector (u_x, u_y, u_z) via Okada (1985) model.
4. Calculate bathymetric slope correction: Delta h = u_z + u_h · grad(b) (Tanioka & Seno 1997).
5. Apply Kajiura water-column filter to obtain initial surface elevation eta(x,y,0).
6. Initialize GPU Numerical Engine (Tsunami-HySEA CUDA Kernels).
7. Dynamically evaluate time step size dt according to CFL condition (CFL <= 0.45).
8. WHILE Simulation Time t < Forecast Horizon DO
9. Compute HLLC numerical fluxes across cell interfaces.
10. Apply TVD MC slope limiter for second-order spatial reconstruction.
11. Enforce well-balanced hydrostatic reconstruction across bed slopes.
12. Update cell depths using positivity-preserving wet/dry algorithms (h >= h_dry).
13. IF DART ocean-bottom pressure data stream is received THEN
14. Perform Ensemble Kalman Filter (EnKF) state analysis update.
15. END IF
16. Save state fields (eta, u, v, d_f) at specified output time intervals.
17. END WHILE
18. Compute summary metrics: Max Flow Depth (d_f), Max Velocity, Arrival Time (T_arr).
19. Export hazard rasters to Cloud-Optimized GeoTIFF (COG) and serve to WebGL frontend.
---

16. Mathematical Appendix and Organized Formula

Reference

Formula 1: Okada Static Vertical Displacement

- **Expression:** u_z(x,y,0) = -\frac{U_2}{2\pi} \left[\frac{q d}{R(R+\xi)} + \sin\delta \arctan\left(\frac{\xi\eta}{q R}\right) - I_5 \sin\delta \cos\delta \right] \Big\vert{}_{\xi = x-L}^{\xi = x} \Big\vert{}_{\eta = p-W}^{\eta = p}
- **Variables:** U_2 (dip slip, \text{m}), \delta (dip angle, \text{rad}), d (fault depth, \text{m}), R = \sqrt{\xi^2 + \eta^2 + q^2} (\text{m}), q = y \sin\delta - d \cos\delta (\text{m}), p = y \cos\delta + d \sin\delta (\text{m}).
- **SI Units:** Surface displacement u_z (\text{m}).
- **Interpretation:** Computes vertical surface deformation due to rectangular fault dislocation in an elastic half-space.
- **Assumptions:** Linear elastic, homogeneous, isotropic half-space.
- **Numerical Notes:** Evaluate arctan branch cuts near q=0 to prevent numerical divergence.

Formula 2: Tanioka-Seno Bathymetric Uplift Adjustment

- **Expression:** \Delta h_{seafloor}(x,y) = u_z(x,y) + u_x(x,y) \frac{\partial b}{\partial x} + u_y(x,y) \frac{\partial b}{\parti[span_252](start_span)[span_252](end_span)[span_254](start_span)[span_254](end_span)[span_256](start_span)[span_256](end_span)al y}
- **Variables:** u_z, u_x, u_y (3D elastic displacement vector, \text{m}), b (bed elevation positive upward, \text{m}).
- **SI Units:** Net seabed elevation change (\text{m}).
- **Interpretation:** Accounts for additional vertical water volume displacement caused by horizontal movement of steep bathymetry.
- **Assumptions:** Rigid seabed surface, small displacement gradients.
- **Numerical Notes:** Central-difference local gradients \nabla b must be smoothed to avoid high-wavenumber noise.

Formula 3: 2D NSWE Mass Conservation

- **Expression:** \frac{\partial h}{\partial t} + \frac{\partial (h u)}{\partial x} + \frac{\partial (h v)}{\partial y} = 0
- **Variables:** h (total depth, \text{m}), u, v (depth-averaged horizontal velocities, \text{m/s}), t (time, \text{s}).
- **SI Units:** Mass conservation rate (\text{m/s}).
- **Interpretation:** Enforces fluid mass conservation across depth-integrated control volume. * **Assumptions:** Incompressible fluid, hydrostatic pressure distribution.
- **Numerical Notes:** Must be solved conservatively to ensure global volume preservation.

Formula 4: 2D NSWE Momentum Conservation

- **Expression:** \frac{\partial (hu)}{\partial t} + \frac{\partial}{\partial x}\left(hu^2 + \frac{1}{2}gh^2\right) + \frac{\partial (huv)}{\partial y} = -gh\frac{\partial b}{\partial x} +

f_{cor}hv - \frac{g n^2 u \sqrt{u^2+v^2}}{h^{1/3}}
- **Variables:** g (9.81 \text{ m/s}^2), b (bed elevation, \text{m}), n (Manning roughness, \text{s}\cdot\text{m}^{-1/3}), f_{cor} (Coriolis parameter, \text{s}^{-1}).
- **SI Units:** Momentum flux density (\text{m}^2/\text{s}^2).
- **Interpretation:** Balances momentum advection, pressure gradients, bed slopes, bottom friction, and Coriolis forces.
- **Assumptions:** Hydrostatic pressure distribution, negligible vertical acceleration.
- **Numerical Notes:** Requires well-balanced reconstruction to prevent artificial motion over sloping topography.

Formula 5: Madsen-Sørensen Boussinesq Dispersion

- **Expression:** \mathbf{\Psi}_{disp} = - \left(B + \frac{1}{3} \right) h_0^3 \nabla \left(\nabla \cdot \frac{\partial \mathbf{u}}{\partial t} \right) - B g h_0^3 \nabla \left(\nabla \cdot (h_0 \nabla \eta) \right)
- **Variables:** B = 1/15 (tuning coefficient), h_0 (still-water depth, \text{m}), \mathbf{u} (velocity vector, \text{m/s}).
- **SI Units:** Dispersive momentum acceleration (\t[span_113](start_span)[span_113](end_span)ext{m}^2/\text{s}^2).
- **Interpretation:** Restores frequency dispersion dynamics to shallow-water models.
- **Assumptions:** Weakly non-linear (\epsilon \ll 1), weakly dispersive (\mu \ll 1).
- **Numerical Notes:** Requires implicit high-order derivative matrix inversions at every time step.

Formula 6: CFL Stability Criterion

- **Expression:** \Delta t \le C_{CFL} \cdot \min_{i,j} \left(\frac{\Delta x}{\vert{}u_{i,j}\vert{} + \sqrt{g h_{i,j}}} \right)
- **Variables:** C_{CFL} \le 0.5, \Delta x (grid size, \text{m}), h_{i,j} (depth, \text{m}), u_{i,j} (velocity, \text{m/s}).
- **SI Units:** Time step (\text{s}).
- **Interpretation:** Establishes explicit numerical time-step limits based on wave phase speed.
- **Assumptions:** Explicit time-stepping scheme.
- **Common Error:** Failing to update \Delta t dynamically during coastal run-up leads to numerical instability.

Formula 7: Regularized Tikhonov Inversion Objective

- **Expression:** J(\boldsymbol{\theta}) = \Vert{}\mathbf{y} - \mathbf{G}\boldsymbol{[span_18](start_span)[span_18](end_span)[span_21](start_span)[span_21](end_span)\theta}\Vert{}_{\mathbf{C}_\varepsilon^{-1}}^2 + \lambda^2 \Vert{}\mathbf{L}\boldsymbol{\theta}\Vert{}^2
- **Variables:** \mathbf{y} (observations), \mathbf{G} (Green's matrix), \boldsymbol{\theta} (fault slip distribution), \mathbf{L} (Laplacian matrix), \lambda (regularization parameter).
- **SI Units:** Dimensionless objective score.
- **Interpretation:** Solves ill-posed source inversions by balancing residual fitting with spatial slip smoothness.

- **Assumptions:** Linear forward wave superposition.
- **Numerical Notes:** Solve using non-negative least squares (NNLS) to enforce positive slip (\boldsymbol{\theta} \ge \mathbf{0}).

Formula 8: Morlet Continuous Wavelet Transform

- **Expression:** W(a, b) = \frac{1}{\sqrt{a}} \int_{-\infty}^{\infty} \eta(t) \pi^{-1/4} e^{i \omega_0 \frac{t-b}{a}} e^{-\frac{1}{2}\left(\frac{t-b}{a}\right)^2} dt
- **Variables:** a (scale parameter), b (time translation, \text{s}), \omega_0 = 6.0 (dimensionless frequency).
- **SI Units:** Waveform coefficient (\text{m}\cdot\text{s}^{1/2}).
- **Interpretation:** Decomposes time-series signal into time-frequency domain to isolate tsunami wave packets.
- **Assumptions:** Stationarity within local Gaussian window.
- **Numerical Notes:** Compute via FFT convolution across discretized scales.

Formula 9: Factorized FNO Layer Operator

- **Expression:** v_{l+1}(x) = \sigma \left(W v_l(x) + \mathcal{F}^{-1} \left(R_{\phi} \cdot \mathcal{F}(v_l) \right)(x) \right)
- **Variables:** \mathcal{F} (Fourier transform), R_{\phi} (learnable complex tensor), W (linear transform), \sigma (activation).
- **SI Units:** Feature map values.
- **Interpretation:** Maps continuous fluid state functions across resolution spaces in Fourier domain.
- **Assumptions:** Periodic boundary extensions during FFT processing.
- **Numerical Notes:** Truncate Fourier modes to k_{max} \le 32 to minimize GPU memory consumption.

Formula 10: Inundation Area Intersection over Union (IoU)

- **Expression:** IoU = \frac{\sum_{i,j}[span_290](start_span)[span_290](end_span)[span_297](start_span)[span_297](end_span) \mathbb{I}\left(d_{sim, ij} > d_{dry} \text{ and } d_{obs, ij} > d_{dry}\right)}{\sum_{i,j} \mathbb{I}\left(d_{sim, ij} > d_{dry} \text{ or } d_{obs, ij} > d_{dry}\right)}
- **Variables:** d_{sim} (simulated flow depth, \text{m}), d_{obs} (observed flow depth, \text{m}), d_{dry} = 0.1\text{ m}.
- **SI Units:** Dimensionless ratio (0.0 \le IoU \le 1.0).
- **Interpretation:** Quantifies spatial accuracy of inundation extent maps against post-event field surveys.
- **Assumptions:** Synchronized spatial raster grid alignment.
- **Numerical Notes:** Highly sensitive to topographic DEM resolution along steep coastlines.

17. Structured Comparative Taxonomy Tables

Table A: Source Mechanism vs. Appropriate Physical Model

| Source Mechanism | Dominant Physics | Governing Model | Input Parameters | Primary

Uncertainty | | :--- | :--- | :--- | :--- | :--- | | **Subduction Megathrust** | Elastic Half-Space | Okada Dislocation | L, W, \theta, \delta, \lambda, U, d | Fault Slip Heterogeneity | | **Strike-Slip (Sloping Bed)** | Horizontal Bed Slope | Tanioka-Seno Uplift | \mathbf{u}_h, \nabla b, L, W, U | Bathymetric Slope Accuracy | | **Submarine Landslide** | Granular Mass Flow | Savage-Hutter / Two-Phase | V_s, \rho_s/\rho_w, \delta_b, Scarp | Slide Kinematics & Volume | | **Volcanic Flank Collapse** | Fluid-Structure Interaction | Multilayer Nonhydrostatic | V_{collapse}, \text{Speed}, \rho_{vol} | Eruption Kinematics | | **Bolide Impact** | Impulsive Cavitation | 3D Compressible Navier-Stokes | \text{Mass}, \text{Velocity}, \text{Impact Angle} | Kinetic Energy Partitioning |

Table B: Governing Equation Family vs. Assumptions and Valid Regimes

Equation System	Primary Assumptions	Validity Regime	Invalid Regimes
Linear Shallow Water (LSWE)	\epsilon \ll 1, \mu \ll 1, Hydrostatic	Deep Ocean Propagation (h > 500\text{m})	Coastal Inundation & Breaking
Nonlinear Shallow Water (NSWE)	\mu \ll 1, Hydrostatic Pressure	Nearshore Shoaling, Inundation	Trans-oceanic Dispersion
Boussinesq Equations	\mu = O(10^{-2}), Weak Nonlinearity	Trans-oceanic Dispersive Propagation	3D Overturning Wave Breaking
Layered Nonhydrostatic	Piecewise Linear w(z)	Submarine Mass Failure Generation	Multi-Phase Sediment Transport
3D Navier-Stokes (VOF)	Continuum Incompressible Flow	Wave Impact on Structures	Basin-Scale Propagation (Cost)

Table C: Numerical Solver vs. Accuracy, Stability, Cost, and Application

| Solver / Scheme | Spatial / Temporal Order | Stability Condition | Computational Cost | Intended Application | | :--- | :--- | :--- | :--- | :--- | | **Finite Volume HLLC (TVD)** | 2nd Order Space / Time | Explicit \text{CFL} \le 0.5 | Moderate (Highly Efficient) | Coastal Inundation & Bores | | **GeoClaw (FV + AMR)** | 2nd Order Space / Time | Explicit \text{CFL} \le 0.45 | Adaptive (CPU Parallel) | Multi-scale Regional Hazard | | *Tsunami-HySEA (FV CUDA)* | 2nd / 3rd Order SSP-RK | Explicit \text{CFL} \le 0.5 | Low (GPU Accelerated FTRT) | Operational Forecasting | | **Discontinuous Galerkin (DG)** | Arbitrary High Order | Strict \text{CFL} \propto (2k+1)^{-1} | Very High | Complex Boundary Benchmarks | | **Boussinesq Finite Difference** | 4th Space / 2nd Time | Implicit Matrix Solver | High | Trans-oceanic Dispersive Waves |

Table D: Inversion Algorithm vs. Data Requirements and Identifiability

Inversion Methodology	Required Sensor Inputs	Computational Latency	Identifiability / Limitations
Regularized Least-Squares	DART, Tide Gauges, GNSS	Low (< 10 \text{ seconds})	Sensitive to Smoothing Parameter \lambda
Bayesian MCMC Inversion	Multi-sensor Networks	High (\approx 10-30 \text{ minutes})	Evaluates Full Posterior Covariance
Ensemble Kalman	Real-time DART	Sequential (< 1	Requires

Inversion Methodology	Required Sensor Inputs	Computational Latency	Identifiability / Limitations
Filter (EnKF)	Streams	\te[span_119](start_span)[span_119](end_span)xt{ s/step})	Representative Ensemble
Adjoint-Based Optimization	Continuous Wavefield State	Moderate-High	Prone to Local Minima Traps

Table E: Machine Learning Method vs. Training, Generalization, and Constraints

ML Architecture	Training Data Requirements	Generalization Bounds	Physical Constraints Imposed
Factorized FNO (F-FNO)	> 10,000 NSWE Simulations	High within Domain Box	Spectral Mode Truncation
Physics-Informed NN (PINN)	PDE Residual Sampling Points	Poor Outside Boundary Box	Soft Loss Regularization
DeepONet Operator	Paired Sensor Input-Output Functions	Moderate Operator Scaling	Soft Physical Penalties
Random Forest Classifier	Historical Event Database	Low (Classification Only)	None (Purely Empirical)

Table F: Dataset vs. Coverage, Resolution, Access, and Validation Value

Dataset Name	Coverage	Spatial / Temporal Res	Access / Licensing	Validation Value
GEBCO Bathymetry	Global Oceans	15 arc-sec	Open / Public Domain	Essential Deep Propagation DEM
NOAA Coastal DEMs	US Coastlines	1/3 - 1 arc-sec	Open REST API	Gold Standard for Coastal Inundation
NOAA DART Array	Global Basins	15s High-frequency	Open NDBC API	Deep Wave Calibration
USGS ComCat	Global Earthquakes	Real-time Alerts	Open GeoJSON API	Instant Source Initialization
IOC Sea Level Network	Global Coastlines	1-minute Readings	Open REST API	Coastal Arrival Time Validation

Table G: Validation Event vs. Available Observations and Measurable Outputs

Validation Event	Primary Tsunami Source	Available Observational Data	Measurable Output Target
Tohoku 2011 (Japan)	M_w 9.0-9.[span_376](start_span)[span_376](end_span)1 Megathrust	GPS Buoys, DART, LiDAR Extent	Run-up > 39\text{m}, Offshore Profiles
Chile 2010 (Maule)	M_w 8.8 Megathrust	DART Array, Tide Gauges	Trans-pacific Amplitude & Timing
Okushiri 1993 (Japan)	M_w 7.7 Complex Fault	High-res Run-up Survey	Monai Valley Local Run-up (31\text{m})
Palu 2018 (Indonesia)	M_w 7.5 Strike-Slip / SMF	CCTV Videos, Field Surveys	Bay Resonance & Local Surge
Hunga Tonga 2022	Volcanic Explosion	Air/Water Pressure Gauges	

Atmospheric Proudman Waves |

Table H: Implementation Feature vs. Importance, Cost, and Priority

Implementation Feature	Scientific Importance	Computational Cost	Development Priority
Well-Balanced FV Solver	CRITICAL (Preserves Lake-at-Rest)	Moderate	Phase 1 (Core Engine)
Positivity-Preserving Wet/Dry	CRITICAL (Prevents Solver Crashes)	Low-Moderate	Phase 1 (Core Engine)
Okada + Tanioka Source	HIGH (Accurate Initial Conditions)	Very Low	Phase 1 (Core Engine)
GPU CUDA Acceleration	HIGH (Enables FTRT Real-Time Runs)	High Dev Effort	Phase 2 (Operational Scale)
Nested Grids / AMR	HIGH (Resolves Coastal Inundation)	High Memory	Phase 2 (Operational Scale)
Dispersive Boussinesq Engine	MEDIUM (Trans-oceanic Waves)	High CPU Cost	Phase 3 (Advanced Physics)
Neural Operator Surrogates	MEDIUM (Rapid Ensemble Sampling)	High Training Cost	Phase 3 (Advanced Physics)

Table I: Recommended Algorithms vs. Alternatives with Justifications

System Module	Recommended Selection	Rejected Alternatives	Evidence-Based Justification
Hydrodynamic Solver	2nd-Order FV (HLLC / TVD)	Standard Finite Difference	Handles Hydrodynamic Bores & Dry Beds
Equilibrium Preservation	Hydrostatic Reconstruction	Unbalanced Source Splitting	Eliminates Spurious Waves over Slopes
Source Kinematics	Okada + Tanioka-Seno Uplift	Pure Vertical Okada Model	Captures Bathymetric Slope Displacements
Compute Engine	GPU Architecture (CUDA/SYCL)	Single-Core CPU Systems	Delivers >100\times Faster-Than-Real-Time
Surrogate Emulator	Factorized FNO (F-FNO)	Standard Dense PINNs	Preserves Mesh Invariance & Scalability

18. Strategic Implementation Roadmap and Concrete Design Recommendations

Capability Level Architecture
LEVEL 1: CREDIBLE PROTOTYPE LEVEL 2: RESEARCH PLATFORM LEVEL 3: OPERATIONAL ARCHITECTURE
[2D NSWE FV Single-Grid] -------> [Multi-GPU Nested-Grid Solver] -------> [Automated Ingestion & EnKF Engine]
- Okada Static Source -
Lands[span_305](start_span)[span_305](end_span)lide-HySEA Integration - F-FNO Physics Sur[span_151](start_span)[span_151](end_span)rogate Emulators
- GEBCO / Local DEM Data - EnKF

Real-Time[span_292](start_span)[span_292](end_span)[span_299](start_span)[span_299](end_span) Assimilation - Probabilistic Hazard Map Generation
- NTHMP Benchmarks Passed - Independent Hindcast Validation - Authoritative Warning Integration

Level 1: Scientifically Credible Prototype

- *Mathematical Physics:* 2D Hydrostatic Non-Linear Shallow-Water Equations (NSWE).
- *Algorithms:* 2nd-Order Well-Balanced Finite Volume Godunov scheme with HLLC Riemann flux, MC TVD slope limiters, and positivity-preserving wetting/drying.
- *Data:* GEBCO 15 arc-second bathymetry combined with NOAA 1 arc-second DEMs.
- *Source Model:* Okada static dislocation combined with Tanioka-Seno bathymetric slope uplift correction.
- *Required Evidence:* Pass NTHMP Benchmarks 1, 4, and 6 with zero mass balance errors.

Level 2: Research-Grade Simulation Platform

* *Mathematical Physics:* Multi-layer non-hydrostatic or Boussinesq extensions coupled with Landslide-HySEA granular mass failure physics.
- *Algorithms:* Multi-GPU parallel execution with nested adaptive mesh refinement (AMR).
- *Data Assimilation:* Ensemble Kalman Filter (EnKF) assimilating real-time DART buoy pressure streams.
- *Required Evidence:* Reproduce 2011 Tohoku and 2010 Maule field inundation observations with IoU \ge 0.70.

Level 3: Operational Forecasting Architecture

- *Architecture:* Low-latency automated ingestion API processing USGS seismic alerts, DART buoys, and tide gauges.
- *Predictive Engine:* Hybrid combination of precomputed scenario database lookups and Faster-Than-Real-Time (FTRT) GPU simulations. * *Surrogate Models:* Factorized Fourier Neural Operator (F-FNO) surrogates generating probabilistic hazard maps via Monte Carlo sampling.
- *Required Evidence:* Formal operational certification by authorized tsunami warning authorities.

Direct Recommendations for Implementation Questions

1. **Which governing equations should be used first?** Use the **2D Hydrostatic Non-Linear Shallow-Water Equations (NSWE)** expressed in conservative vector form. They provide the optimal balance between computational speed and physical accuracy across coastal shoaling and inland inundation.
2. **Which numerical solver is the best starting point and why?** Implement a **2nd-Order Godunov-Type Finite Volume Solver with an HLLC Riemann Solver** and **Hydrostatic Reconstruction**. This preserves the lake-at-rest state unconditionally over non-flat bathymetry and handles wetting/drying transitions stably.
3. **Which source-estimation method should be implemented first?** Implement a

Regularized Linear Least-Squares Sub-Fault Slip Inversion with non-negativity constraints (\boldsymbol[span_219](start_span)[span_219](end_span)[span_224](start_span)[span_224](end_span)[span_229](start_span)[span_229](end_span){\theta} \ge \mathbf{0}) using precomputed Green's function matrices.

4. **Which datasets are essential?** Ingest **GEBCO 2024** for ocean bathymetry, *NOAA NCEI DEMs* for coastal elevation, and the **NOAA DART Buoy API** for ocean wave verification.
5. **Which benchmark events should be used?** Validate against **NTHMP Benchmark 1** (Planar Beach Run-up), **NTHMP Benchmark 6** (Okushiri Island Inundation), and the **2011 Tohoku Earthquake** field dataset.
6. **Which outputs can be claimed scientifically?** Claim deep-ocean travel times (\pm 2 \text{ min}), offshore DART wave amplitudes, and regional inundation extents (IoU \ge 0.70) when supported by high-resolution (< 10\text{m}) coastal LiDAR DEMs.
7. **Which claims must be prohibited until validated?** Prohibit claims of exact localized structural flow depths or site-specific building damage probabilities when using coarse global bathymetry (> 30\text{m}) without local friction calibration.
8. **What should be built first, second, and third?** *First:* The 2D Finite Volume Hydrostatic Reconstruction Solver Core. *Second:* The Okada and Tanioka-Seno source initialization modules. *Third:* The GPU CUDA acceleration framework and real-time DART data assimilation pipeline.
9. **What is the smallest meaningful experiment that can falsify the chosen approach?** Run the solver over a domain containing a steep submerged ridge with zero initial velocity and constant surface elevation (\eta = 0). If the scheme generates artificial velocity spikes (\vert{}\mathbf{u}\vert{} > 10^{-6} \text{ m/s}), the well-balanced property is violated, rendering the solver unsuitable for coastal inundation modeling.
10. **What evidence would justify claiming that the resulting system works?** Achieving passing scores across all **NTHMP Benchmarks (BP1 through BP6)**, maintaining mass conservation error below 10^{-12}, and demonstrating multi-GPU execution speeds greater than 100\times faster than real-time on standard operational benchmark domains.

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